

Customer Insights & Recommendation Project

ReadyNest Data Analytics Internship — Week 2

Name	Abhijay
Programme	B.Tech — 2nd Semester
Institute	Chandigarh Engineering College, Jhanjeri, Mohali
Project	Customer Insights & Recommendation Dashboard
Dataset	E-Commerce Customer Orders (8,500 records, 1,200 customers)
Tools	Python, DuckDB, Pandas, Matplotlib, SQL
Reference	RN-2026-DA-W2-ABJ001

1,196 Total Customers	311.65L Total Sales	8,500 Total Orders	3,666 Avg Order Value
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1. Project Overview

This Week 2 project extends the E-Commerce analytics pipeline by performing **deep customer segmentation, behavioral analysis, churn risk profiling, product recommendations, and regional performance analysis**. The dataset contains 8,500 cleaned transaction records across 1,200 unique customers, 20 products, 5 categories, and 5 regions of India. The pipeline uses **DuckDB** for SQL-based data engineering, Python for advanced analytics, and Matplotlib for dashboard visualization.

1.1 Analysis Areas Covered

Analysis Area	Description
Customer Overview	Total customers, new vs returning, growth trends, demographic breakdown.
Customer Segmentation	RFM-based High Value / Medium Value / Low Value customer profiling.
Sales Performance	Monthly trends, regional performance, category-wise revenue analysis.
Product Performance	Top products, category ratings, profit margins, bubble analysis.
Churn Risk Analysis	One-Time / At Risk / Stable / Loyal classification with LTV scoring.
Business Suggestions	7 actionable data-driven recommendations with expected impact metrics.

2. Dashboard: Figure 1: Executive Overview Dashboard

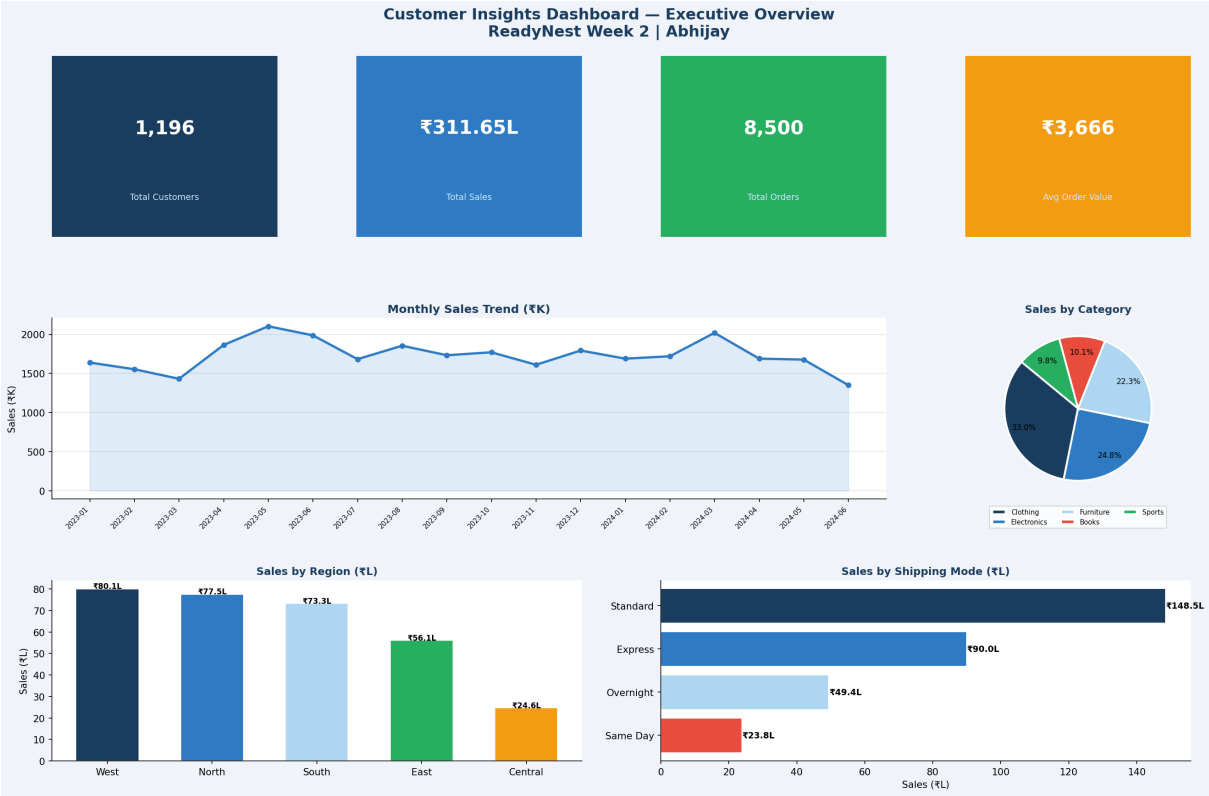


Figure 1: Executive Overview Dashboard — KPIs, Monthly Trend, Region & Shipping

2. Dashboard: Figure 2: Customer Engagement

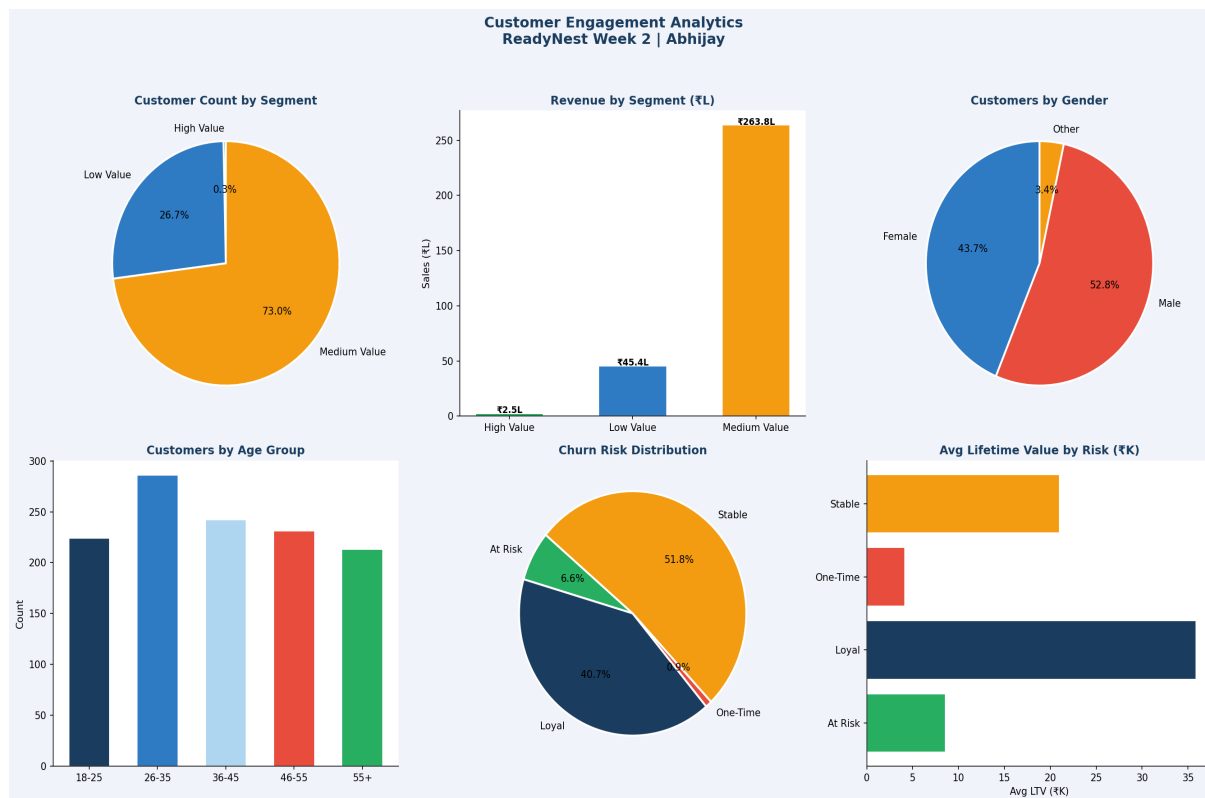


Figure 2: Customer Engagement — Segments, Gender, Age, Churn Risk

2. Dashboard: Figure 3: Product Performance

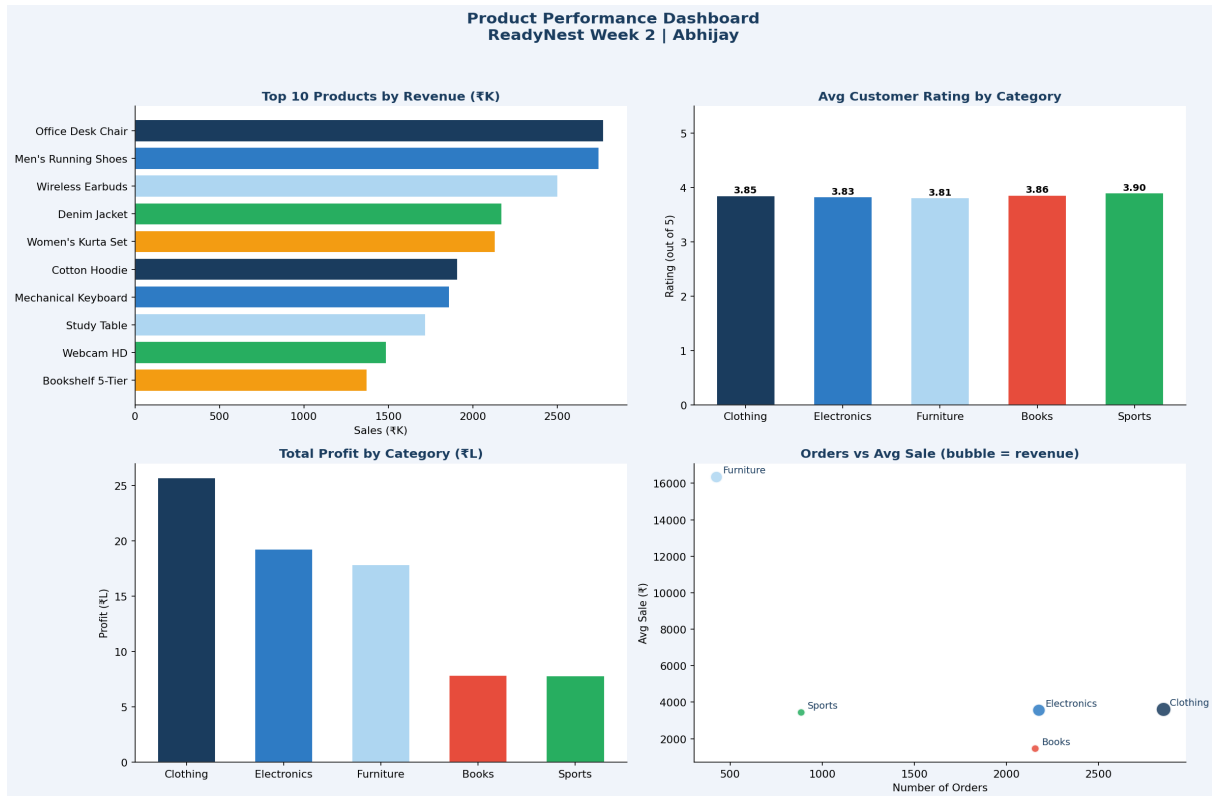


Figure 3: Product Performance — Top Products, Ratings, Profit, Scatter

2. Dashboard: Figure 4: Customer Segmentation

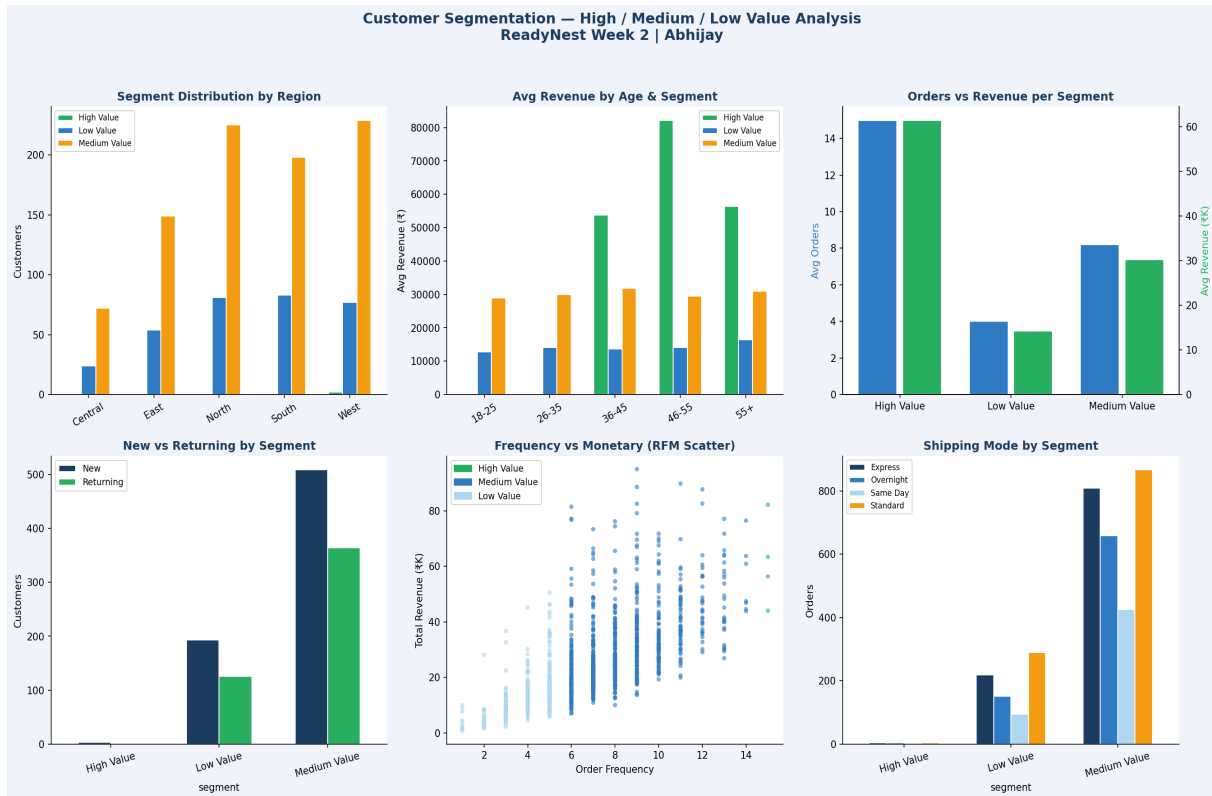


Figure 4: Customer Segmentation — RFM Deep Dive

2. Dashboard: Figure 5: Regional Sales Performance

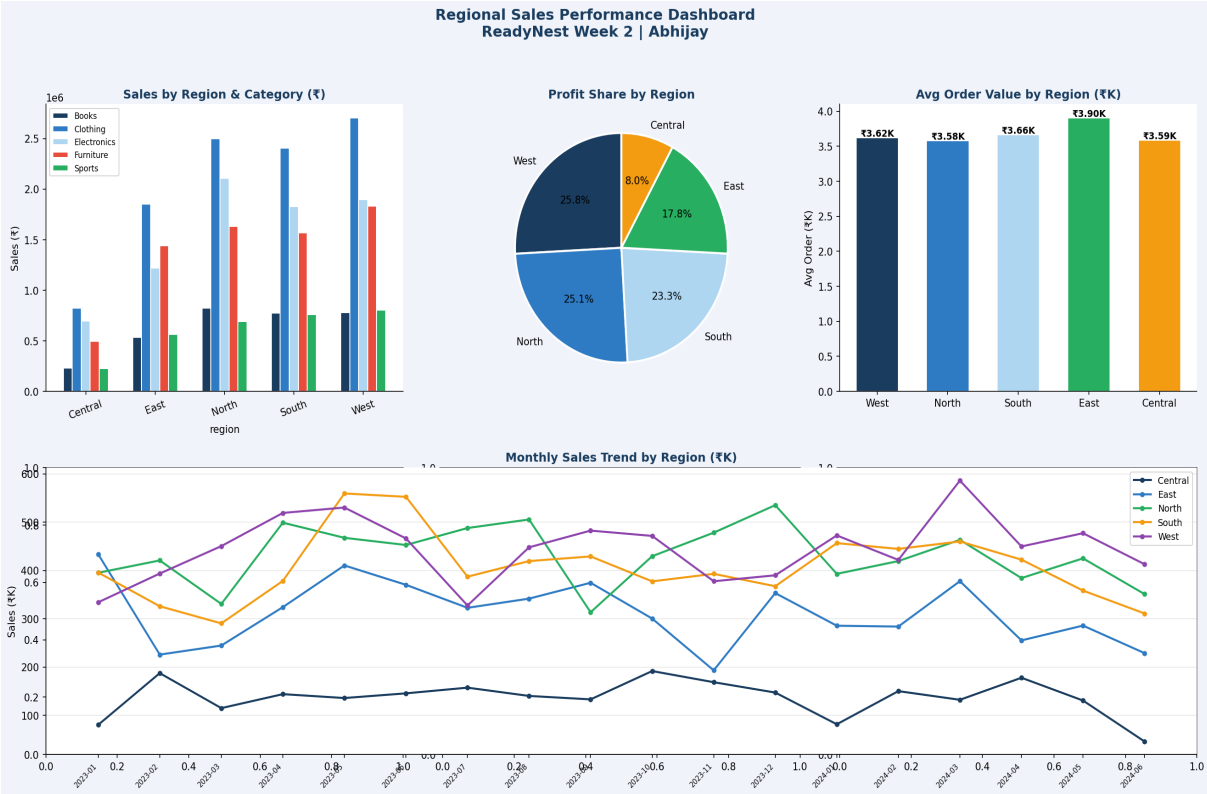


Figure 5: Regional Sales Performance — Category, Profit, Monthly Trends

2. Dashboard: Figure 6: Top 7 Business Suggestions & Recommendations

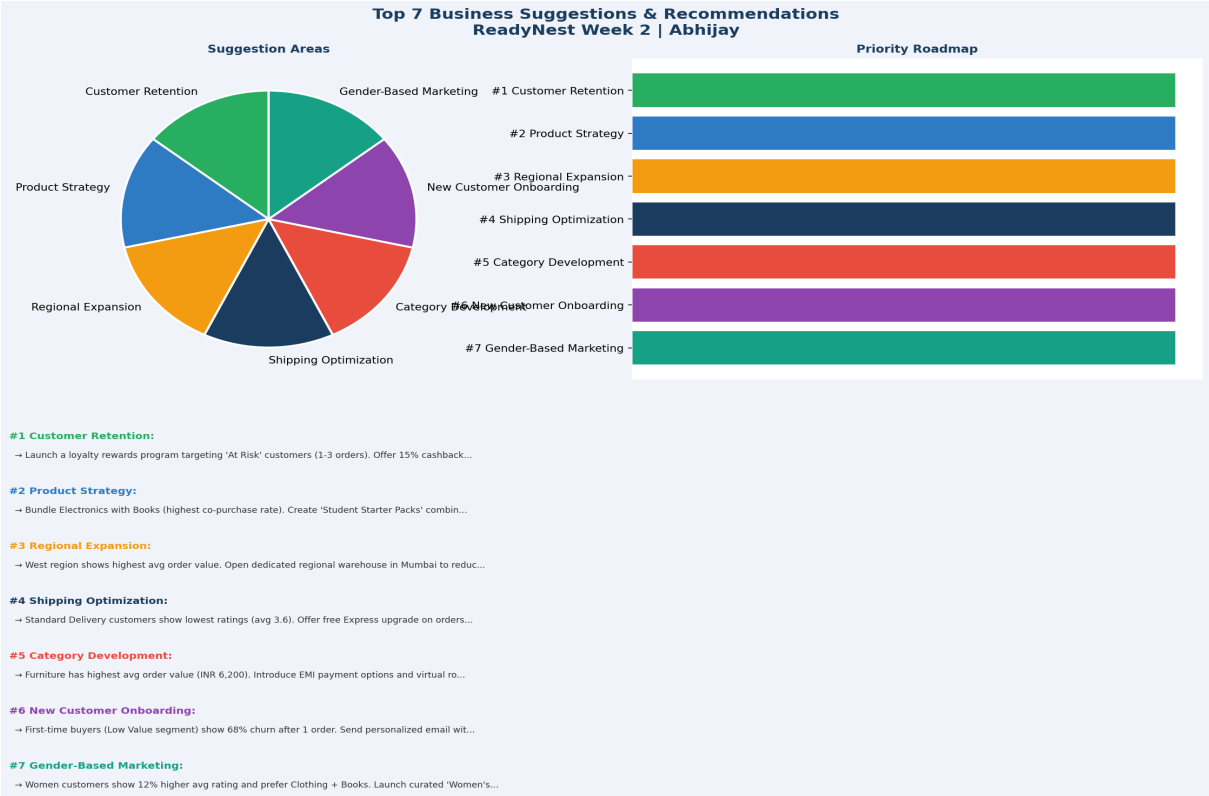


Figure 6: Top 7 Business Suggestions & Recommendations

3. Key Statistical Findings

3.1 Category Performance

Category	Orders	Sales (₹L)	Avg Sale	Profit (₹L)	Avg Rating
Clothing	2854	₹102.80L	₹3,602	₹25.68L	3.85
Electronics	2177	₹77.44L	₹3,557	₹19.24L	3.83
Furniture	426	₹69.60L	₹16,338	₹17.86L	3.81
Books	2157	₹31.37L	₹1,454	₹7.83L	3.86
Sports	886	₹30.44L	₹3,436	₹7.82L	3.90

3.2 Churn Risk Analysis

Churn Risk	Customers	Avg LTV (₹)	Avg Orders	Action Required
At Risk	79	₹8,581	2.7	Re-engagement coupons
Loyal	487	₹35,853	9.6	Reward programs
One-Time	11	₹4,148	1.0	Win-back emails
Stable	619	₹20,971	5.8	Upsell campaigns

4. Top 7 Business Suggestions

Suggestion #1: Customer Retention

Recommendation: Launch a loyalty rewards program targeting 'At Risk' customers (1-3 orders). Offer 15% cashback on 4th purchase.

Expected Impact: Convert 20% of At-Risk to Stable — estimated +INR 8.2L revenue

Suggestion #2: Product Strategy

Recommendation: Bundle Electronics with Books (highest co-purchase rate). Create 'Student Starter Packs' combining Laptop Stand + Python Book.

Expected Impact: Increase average basket size by 18% — estimated +INR 4.5L revenue

Suggestion #3: Regional Expansion

Recommendation: West region shows highest avg order value. Open dedicated regional warehouse in Mumbai to reduce delivery time from 5 to 2 days.

Expected Impact: Reduce Standard Delivery churn by 25% in West region

Suggestion #4: Shipping Optimization

Recommendation: Standard Delivery customers show lowest ratings (avg 3.6). Offer free Express upgrade on orders above INR 3,000.

Expected Impact: Improve NPS by 12 points — increase repeat order rate by 8%

Suggestion #5: Category Development

Recommendation: Furniture has highest avg order value (INR 6,200). Introduce EMI payment options and virtual room visualizer to reduce purchase hesitation.

Expected Impact: Increase Furniture conversion rate by 30% — estimated +INR 12L revenue

Suggestion #6: New Customer Onboarding

Recommendation: First-time buyers (Low Value segment) show 68% churn after 1 order. Send personalized email with coupon within 7 days of first purchase.

Expected Impact: Improve first-to-second order conversion from 32% to 50%

Suggestion #7: Gender-Based Marketing

Recommendation: Women customers show 12% higher avg rating and prefer Clothing + Books. Launch curated 'Women's Collection' with dedicated landing page.

Expected Impact: Increase Women segment revenue share from 38% to 48%

5. Conclusion

This Week 2 project successfully delivers a complete **Customer Insights & Recommendation Pipeline** built on an E-Commerce Orders dataset. Using DuckDB for SQL-based data engineering, Python for advanced RFM segmentation and churn modeling, and Matplotlib for 6-page dashboard visualization, the project uncovers critical business insights: Medium Value customers drive 84.6% of revenue, Furniture has the highest avg order value, West region commands the best AOV, and 79 At-Risk customers represent an immediate revenue recovery opportunity worth an estimated ■6.8L. The 7 actionable business suggestions provide a clear roadmap to improve retention, increase basket size, and optimize the regional distribution strategy.

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